

GBM8360E - Physical Principles of Advanced MRI

Laboratory 2 (A2026)

Polytechnique Montréal

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PLEASE READ THE FOLLOWING CAREFULLY.

- This lab accounts for **6% of your final course grade**.
- Please hand in your completed work before **Wednesday, September 30, 12:45 p.m.**, before the start of the next laboratory. If this poses a problem for good reasons (illness or a scheduling conflict, for example), please contact the instructor or TA.
- This is a take-home lab, and everyone needs to submit their own work. You are entitled to consult anyone at all for general explanations, including your classmates in this course or your instructor. However, you are not entitled to consult your classmates or anyone else in order to copy the complete answers or parts of the answers, and you are not entitled to let yourself be consulted by classmates for that purpose. A departure from this rule will be treated as plagiarism as defined by the regulations of Polytechnique Montréal. What this means is that you have the right (and indeed the duty) to learn, to understand, to help, to be helped, to explain and to have things explained to you; but you do not have the right to copy someone else's answers mindlessly, nor to let someone else copy yours.
- You are entitled to your notes, your books, the internet, and so on.
- LLM usage is allowed except when explicitly discouraged in the description of the exercise. In general, LLMs should not be used to answer any question that asks for your observations or intuition. These questions are not graded as right or wrong; they are meant to build your own intuition for the material. Using an LLM here only cheats you out of that.
- Please cite all references other than the course notes (people, books, internet, etc.) at the point where they apply.
- Read the complete document before you begin.
- You are responsible for your own work. Keep multiple backups (locally and online).

Grading scheme

Component	Points
MyST book and formatting	/ 7
Interactive figures and formatting	/ 6
MyST/Markdown formatted explanations	/ 4
Quality* * One ranking per book, decided by the grader(s) on insight beyond the static Lab 1 figures, clarity, and originality. Tied books may share a band. Missing or incomplete explanations receive 0.	Ranked: • 1st and 2nd place = 3 points • 3rd and 4th = 2 points • 5th and 6th = 1 point
Total	/ 20

Guide to submitting Lab 2

This lab is a follow-up to Exercise 3 of Lab 1.

The purpose of this lab is to:

- learn the tools used to create the Quantitative MRI mOOC book, which you will need for your final project (e.g. MyST, Jupyter Notebook, Markdown, GitHub, Plotly, etc).
- apply these tools (in particular, the interactive component) in order to better explain and better understand the phenomena explored in Exercise 3 of Lab 1

What matters in this lab is: understanding the concepts and being able to interpret your results, rather than simply obtaining the expected output.

On Moodle, submit two links, and nothing else:

1. your repository: [<your-github-username>](https://github.com/GBM8360/lab2-)
2. your published book: [<your-github-username>](https://gbm8360.github.io/lab2-)

Your book is graded from the published website and the repository. The GitHub Actions build of your last commit before the deadline must pass.

General instructions are provided below:

- Don't link to externally hosted files on services like Google Drive and OneDrive, as these are often not configured with the correct permissions to be readable. An unreachable file cannot be graded.
- Qualitative MRI images (eg, T1w, T2w) can be shown in greyscale, but the brightness and contrast should be adjusted so that the main structures are visible. Quantitative MRI images as well as phase images should be shown in colour with colour bars.
- Images whose values are symmetric about a meaningful centre (phase from $-\pi$ to π , or B_1 from 0.7 to 1.3, where 1 is ideal) should use a diverging colourmap that visibly separates low values, high values, and the midpoint. Blue-white-red is a common choice.
- You must share your code and/or data; for code, a GitHub repo is the preferred way. For data, use either a public link (eg, Zenodo) or a file attached to a GitHub release (these are not version controlled and do not get downloaded with each clone of the repo).
- Everything must be grouped together with its relevant text (e.g. like a book: graphs, data tables, code listings, drawings...). That is, don't add all your images at the end of the file and refer to them – add them inline with your text.

Good luck!

EXERCISE.

This lab is designed to help you learn the tools you will need for the final project of this course (an interactive MyST contribution to the [mOOC book](#) and/or a new one on more general MRI imaging principles).

Using the supplied MyST book template (see resources section below), create a MyST book that re-explores the three manipulations done on raw k-space data in Exercise 3 of Lab 1. As a reminder, here were the three parts:

1. Mask the central region of k-space (e.g. using $\frac{1}{2}$ of the number of voxels in both dimensions). Display the resulting k-space and magnitude of the image.
2. Downsample the k-space by half in one direction (e.g., use any tool or method you want). Display the resulting k-space and magnitude of the image.
3. Assume a Cartesian acquisition scheme. At some point during the acquisition (± 5 lines near the center, your choice), imagine the patient rotated their head suddenly by 20 degrees and then stayed still. Manipulate the k-space (and if needed, image space) data to simulate this.

The MyST book should contain one page (Markdown file) per part listed above. For each page, leverage the interactivity enabled by tools like Plotly to illustrate the phenomena beyond what a simple PDF or book is capable of doing. Meaning, explore visually how k-space and image space change if you vary any related variable (e.g. for the rotation, this could be the k-space position when the movement happens, or the angle of rotation). Try changing different things before deciding what you want to explain and how; the purpose here is for you to either discover something you hadn't the previous week using the static images, or to deepen your understanding and then communicate it in a way that could only be explained well using this interactivity. Beyond explaining this, you are encouraged to also speculate / ask future questions about things that this revealed (i.e. could you "fix" this retrospectively on this dataset? If so, how? Would a retrospective fix work if this phenomenon happened in a real experiment? why or why not. etc.)

Note: If you don't find something interesting in one or more of the examples above, you have some allowance to do something different but related. E.g. masking a circular region, zero-filling instead of just downsampling, changing the type of motion, etc.

RECOMMENDED FEATURES

The book should try and leverage the tools and features made available by the MyST format. As such, we strongly recommend doing (at least) the following:

- Write an introduction page (index.md) explaining what the book is.
- Put each part on its own page, meaning a separate Markdown file listed in the toc of myst.yml. At minimum, the book has an introduction and three pages (four Markdown files).
- Customizing the myst.yml properties
- Using all general Markdown syntax as needed (e.g., links, tables, etc)
- Embedding the interactive figures into the page via their labels.
- Displaying equations via LaTeX in the book
- Using the MyST cross-referencing syntax when mentioning figures, tables, or equations in text.
- Adding references via a BibTeX bibliography in the MyST book and citing them using the `{cite:p}` syntax.
- Using code blocks and admonitions as necessary.
- Adding acronyms and abbreviation explanations using the MyST in-line syntax or frontmatter.

and more that may not be listed here. Use your judgement, read the documentation, or ask for help.

GETTING STARTED

To help you get started with MyST, we've created a template repo that once set up, will build a MyST website via GitHub pages at each commit on GitHub via GitHub actions.

1. Send your GitHub username to the TA and accept the invitation to the GBM8360 organization.
 - a. Should be done prior to the lab.

2. Open the template, click **Use this template**
3. Repository setup:
 - a. Owner: **GBM8360 organization**
 - b. Naming convention: **lab2-<your-github-username>**
 - c. Visibility: **Public**
4. The first build will fail. This is expected.
 - a. Go to Settings → Pages → Build and deployment → Source → select GitHub Actions
 - b. Then, go to Actions → the failed run → **Re-run failed jobs**. From then on, every push rebuilds and republishes your book.
5. To work locally, follow the template's README.md (pip install -r requirements.txt, then myst start for a live preview).
6. Put the data your notebooks read inside your repository OR fetch it during the build via the release link
7. Add every Python package you import (for example scipy) to requirements.txt, or the build will fail.

RESOURCES

MyST book template: <https://github.com/GBM8360/lab2-myst-template>

MyST documentation: <https://mystmd.org/guide/>

Plotly.py documentation: <https://plotly.com/python/>

Dataset link: https://github.com/GBM8360/laboratory_1/releases/tag/v0.1

Filename: subject-02_raw_kspace.zip